Autonomous Data Scientist: An Agent-Based Platform for Automated End-to-End Data Analysis, Machine Learning and Evidence-Based Report Generation

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Abstract:Manual data-science workflows require analysts to repeatedly perform dataset inspection, data cleaning, exploratory analysis, visualization, model selection, interpretation and report preparation. This process is time-consuming, repetitive and dependent on the analyst's technical experience. This paper presents Autonomous Data Scientist, an agent-based web platform that automates the end-to-end data-science workflow from raw dataset upload to final analytical report generation. The system decomposes the workflow into specialised agents for data profiling, cleaning, exploratory data analysis (EDA), machine-learning modeling, visualization and AI-powered insight generation. A central orchestration layer executes these agents sequentially, maintains intermediate analytical results and passes validated outputs between stages. Users upload a raw structured dataset, after which the Profiling Agent identifies schema, data types, missing values, duplicates and statistical characteristics. The Cleaning Agent performs appropriate preprocessing, while the EDA Agent automatically analyses distributions, relationships and correlations. Where predictive analysis is applicable, the Modeling Agent prepares features, trains candidate models and evaluates them using suitable metrics. The Visualization Agent generates charts based on dataset characteristics, and the AI Insights Agent converts computed analytical results into evidence-based natural-language observations. An AI Chat module allows users to interact with the generated analytical context, while a final Report module consolidates the outputs into an executive summary, data-quality assessment, EDA findings, model results, visualizations, insights and recommendations. The platform demonstrates how agentic AI can automate repetitive analytical tasks while preserving intermediate results for transparency and inspection. Functional testing showed successful execution of the complete workflow across representative tabular datasets. The main limitations are automated target identification, dependence on dataset quality, potential AI-generated unsupported interpretations and limited support for non-tabular data. Future work includes multi-agent collaboration, autonomous feature engineering, model explainability, statistical validation, experiment tracking and human-in-the-loop approval mechanisms.
Index Terms— autonomous data scientist, agentic AI, automated data analysis, multi-agent systems, exploratory data analysis, automated machine learning, data cleaning, data visualization, machine learning, AI insights.

INTRODUCTION

Data analysis is an essential component of modern decision-making across business, healthcare, finance, education and scientific research. Although numerous libraries and platforms simplify individual data-science tasks, analysts still need to manually coordinate multiple stages of the analytical workflow. A typical process involves loading a dataset, understanding its structure, identifying data-quality problems, cleaning the data, performing exploratory analysis, selecting appropriate visualizations, training machine-learning models and preparing a final report.

For users without extensive data-science expertise, determining the correct sequence of these operations can be difficult. Even experienced analysts spend significant time performing repetitive tasks such as checking missing values, identifying duplicate records, calculating descriptive statistics, generating standard visualizations and comparing baseline models.

Three gaps make conventional data-analysis workflows inefficient. First, analytical operations are distributed across multiple tools and libraries, requiring users to manually coordinate the workflow. Second, individual automation tools generally address only one stage, such as visualization or machine learning, rather than the complete lifecycle. Third, analytical findings often need to be manually translated into natural-language explanations and reports.

Recent developments in artificial intelligence and agent-based systems provide an opportunity to automate the coordination of these tasks. Instead of treating data analysis as a collection of independent operations, an autonomous data scientist can observe the characteristics of a dataset, determine appropriate analytical operations, execute specialised tools and combine their outputs.

This paper presents **Autonomous Data Scientist**, a web-based agentic platform that automates the complete

analytical workflow from raw dataset ingestion to final report generation. Its contributions are:

1. an agent-based architecture that decomposes the data-science lifecycle into specialised analytical agents;
2. automated dataset profiling and data-quality assessment;
3. automated data cleaning and preprocessing;
4. autonomous exploratory data analysis and visualization selection;
5. automated machine-learning model training and evaluation when predictive analysis is applicable;
6. AI-generated, evidence-based insights grounded in computed analytical results;
7. an interactive AI Chat interface for querying the analysed dataset; and
8. automated generation of a consolidated final analytical report.

The rest of the paper is organised as follows. Section II reviews background and existing approaches, Section III describes the system architecture, Section IV explains the methodology and agent algorithms, Section V covers implementation, Section VI presents testing and results, and Section VII discusses limitations and future work.

II. BACKGROUND AND RELATED WORK

A. Enabling Concepts

Agentic AI provides a framework in which software agents can observe information, perform reasoning or decision-making, invoke tools and produce outputs toward a specified objective. In the proposed system, individual agents are assigned specialised data-science responsibilities.

Exploratory Data Analysis provides statistical and visual techniques for understanding distributions, relationships, outliers and patterns before formal modeling.

Automated Machine Learning (AutoML) automates parts of model development, including preprocessing, algorithm selection and hyperparameter optimization. However, conventional AutoML generally focuses on predictive modeling rather than the complete analytical workflow.

Large Language Models (LLMs) can transform structured analytical results into natural-language explanations and provide conversational interfaces for interacting with data. In this system, AI-generated explanations are grounded in outputs produced by analytical tools.

Data visualization provides graphical representations of distributions, relationships and model performance, allowing analytical findings to be interpreted more easily.

B. Existing Approaches

Existing data-analysis tools generally fall into several categories.

Notebook-based systems provide extensive flexibility but require the analyst to manually write and execute analysis code.

Business-intelligence platforms provide dashboards and visualization capabilities but commonly require users to manually configure analytical workflows.

AutoML systems automate model development but may not automatically perform complete data profiling, exploratory analysis, insight generation and report creation.

LLM-based data-analysis assistants can interpret natural-language questions and generate analysis, but their reliability depends on tool integration, execution context and grounding of generated statements in actual computed results.

C. Limitations of Current Systems

No integrated workflow. Many tools focus on individual stages such as visualization, cleaning or modeling rather than the complete data-science lifecycle.

High manual effort. Users must often determine which analytical operation should be performed next.

Limited automated reporting. Analytical results frequently require manual interpretation and documentation.

Tool fragmentation. Analysts may need to move between data-processing libraries, visualization software, machine-learning frameworks and reporting tools.

Limited autonomous reasoning. Traditional pipelines execute predefined operations but do not dynamically coordinate specialised analytical agents according to dataset characteristics.

Autonomous Data Scientist is designed to address these limitations through a unified agent-based workflow.

III. SYSTEM ARCHITECTURE

Autonomous Data Scientist uses a modular client-server architecture. A web frontend provides interfaces for dataset upload, profiling, cleaning, EDA, modeling, visualization, AI insights, conversational analysis and report generation.

The backend contains an orchestration layer that coordinates specialised data-science agents. Each agent receives the current dataset or analytical context, performs

its assigned operation and stores its output for subsequent agents.

TABLE I. TECHNOLOGY STACK

Layer	Technology and role
Frontend	React, responsive web interface
Build Tool	Vite
Styling	TailwindCSS
Backend	Python-based API/application server
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn and compatible ML libraries
Visualization	Matplotlib / Plotly
AI Layer	Large Language Model / AI service
Database	Persistent storage for datasets and analysis metadata
Deployment	Containerised/cloud deployment

B. Functional Modules

Dataset Upload. Users upload raw structured datasets through the web interface.

Profiling Agent. Analyses schema, data types, missing values, duplicates, cardinality and descriptive statistics.

Cleaning Agent. Detects and resolves data-quality issues and prepares the dataset for downstream analysis.

EDA Agent. Automatically analyses distributions, correlations, relationships and statistical patterns.

Modeling Agent. Determines whether predictive modeling is applicable, prepares features, trains candidate models and evaluates their performance.

Visualization Agent. Selects and generates appropriate visual representations of the dataset and analytical results.

AI Insights Agent. Converts computed results into evidence-based natural-language insights.

AI Chat. Allows users to ask questions about the uploaded dataset and generated analysis.

Report Generator. Consolidates all agent outputs into a structured analytical report.

C. Data Model

The system maintains an analytical context centred on the uploaded dataset. A dataset record contains metadata such as:

- filename;
- number of rows and columns;
- column names;
- detected data types;
- profiling results;
- cleaning operations;
- EDA results;
- model results;
- generated visualizations;
- AI insights; and
- report-generation status.

Each agent contributes outputs to the shared analytical context.

D. Role-Based Interfaces

The primary interface is designed around the analytical workflow.

Users can access:

- **Overview** for overall analysis status;
- **Upload Dataset** for raw data ingestion;
- **Profiling** for structural and quality analysis;
- **Cleaning** for preprocessing results;
- **EDA** for exploratory findings;
- **Modeling** for machine-learning results;
- **Visualization** for generated charts;
- **AI Insights** for interpreted findings;
- **AI Chat** for interactive analysis; and
- **Report** for the final consolidated output.

The system can also enforce application-level access control when deployed in a multi-user environment.

IV. METHODOLOGY

A. Development Process

The project follows an iterative development process consisting of the following stages:

Dataset ingestion and application interface;

Profiling and data-quality analysis;

Cleaning and preprocessing;

Exploratory data analysis;

Machine-learning integration;

Visualization generation;

AI insight generation;

AI conversational interface;

Automated report generation; and

End-to-end testing and refinement.

Each stage is implemented as an independent module or agent so that the system can be extended without redesigning the entire application.

B. Autonomous Data-Science Pipeline

A dataset $D$is submitted to the orchestration layer. The system first profiles $D$and determines its structural characteristics.

The pipeline then proceeds through data cleaning and exploratory analysis. If the dataset contains a suitable predictive target, the Modeling Agent performs machine-learning analysis. Visualization and insight-generation stages operate on the outputs of previous agents.

Algorithm 1: Autonomous Data-Science Workflow

Input: Raw dataset D

Load dataset D

Validate file format and schema

Execute Profiling Agent

Store profiling results

Execute Cleaning Agent

Validate cleaned dataset

Execute EDA Agent

Store exploratory results

Determine whether predictive modeling is applicable

If applicable:

identify target variable;

prepare features;

train candidate models;

evaluate models;

Execute Visualization Agent

Generate analytical charts

Execute AI Insights Agent

Validate and store generated insights

Generate executive summary

Generate final report

Make analytical context available to AI Chat

Output: Analytical results, visualizations, insights and final report.

C. Data Profiling

The Profiling Agent identifies:

number of records;

number of variables;

numerical variables;

categorical variables;

date/time variables;

missing values;

duplicate records;

unique-value counts;

statistical summaries; and

potential target variables.

For a numerical variable X, the mean is calculated as:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

and the standard deviation is calculated as:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

These statistics provide an initial understanding of the dataset.

D. Automated Data Cleaning

The Cleaning Agent analyses identified data-quality issues and performs appropriate preprocessing operations.

Possible operations include:

missing-value handling;

duplicate removal;

data-type conversion;

categorical normalization;

outlier detection;

categorical encoding;

numerical scaling; and

invalid-value handling.

The agent records the transformations performed so that they can be presented in the final report.

E. Exploratory Data Analysis

The EDA Agent automatically selects analyses based on variable characteristics.

For numerical variables, it can calculate:

mean;

median;

standard deviation;

minimum;

maximum;

quartiles; and

correlation.

For categorical variables, it can analyse:

frequency;

cardinality;

dominant categories; and

category distributions.

For two numerical variables, Pearson correlation is calculated as:

$$r = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2 \sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

The resulting statistics are passed to the Visualization and AI Insights Agents.

F. Automated Modeling

When a suitable target variable is available, the Modeling Agent determines the likely prediction problem.

Classification problems may use metrics such as:

accuracy;

precision;

recall;

F1-score; and

ROC-AUC.

Regression problems may use:

MAE;

MSE;

RMSE; and

R^2.

Candidate models are trained using the prepared dataset and evaluated using appropriate validation procedures.

The system records model metrics and makes them available to the report generator and AI Chat.

G. Visualization Generation

The Visualization Agent selects charts according to data characteristics.

Examples include:

Analytical requirement	Visualization
Numerical distribution	Histogram
Outlier analysis	Box plot
Category frequency	Bar chart
Numerical relationship	Scatter plot
Correlation analysis	Heatmap
Time-dependent variable	Line chart
Model comparison	Performance chart

The generated visualizations are stored as part of the analytical context.

H. AI Insight Generation

The AI Insights Agent receives structured outputs from the analytical modules rather than relying only on raw user input.

The workflow is:

Computed Result → Analytical Context → AI Interpretation → Evidence-Based Insight

For example, if analysis identifies a strong statistical relationship between two variables, the AI agent can describe the observed association while avoiding unsupported causal claims.

This grounding mechanism is important because AI-generated text should remain connected to measurable results produced by the analytical pipeline.

V. IMPLEMENTATION

The frontend provides a dashboard containing navigation options for Overview, Upload Dataset, Profiling, Cleaning, EDA, Modeling, Visualization, AI Insights, AI Chat and R

DA, Modeling, Visualization, AI Insights, AI Chat and Report.

The backend processes the uploaded dataset and maintains the state of each analytical stage.

The data-processing layer uses tabular data-processing libraries for:

- reading datasets;
- type detection;
- statistical computation;
- missing-value analysis;
- duplicate detection;
- preprocessing; and
- transformation.

The machine-learning layer provides model training and evaluation.

The visualization layer generates analytical charts based on the results.

The AI layer receives structured analytical context and generates explanations and insights.

The report layer combines the outputs into a single report.

A. Agent Orchestration

The orchestration component is responsible for:

- initiating agents;
- passing datasets between agents;
- storing intermediate outputs;
- checking stage completion;
- handling errors; and
- triggering subsequent stages.

For example, the EDA Agent does not need to independently determine whether the uploaded file is valid. It receives the validated dataset produced by earlier stages.

This separation makes the architecture modular.

B. AI Chat

The AI Chat module provides an interactive interface over the generated analytical context.

A user can ask:

"What are the major findings from this dataset?"

or:

"Which variables contain missing values?"

or:

"Explain the model results."

The system uses the stored analytical outputs to formulate its response.

C. Final Report

The Report module combines all outputs into an evidence-led analytical document.

The report contains:

1. Executive Summary
2. Dataset Overview
3. Data Quality
4. Cleaning Summary
5. Exploratory Data Analysis
6. Modeling Results
7. Visualizations
8. AI Insights
9. Recommendations

This allows the user to move from raw data to a complete analytical deliverable without manually preparing each section.

VI. TESTING AND RESULTS

A. Strategy and Environment

Testing consists of unit tests for analytical functions, integration tests for the individual agents and end-to-end tests covering the complete dataset-to-report workflow.

Representative structured datasets are used to evaluate:

- file ingestion;
- profiling;
- missing-value detection;
- duplicate detection;
- cleaning;
- EDA;
- visualization;
- model training;
- AI insight generation; and
- report generation.

B. Test Cases

TABLE II. FUNCTIONAL TEST CASES

ID	Scenario	Expected Outcome
1	Upload valid CSV dataset	Dataset successfully loaded
2	Dataset contains missing values	Missing values identified
3	Dataset contains duplicate rows	Duplicate records detected
4	Dataset contains numerical variables	Statistical summary generated
5	Dataset contains categorical variables	Category distributions generated
6	Dataset contains suitable prediction target	Candidate models trained and evaluated
7	Dataset has no suitable prediction target	EDA completed without forced modeling
8	Visualization stage executed	Appropriate charts generated
9	AI Insights stage executed	Analytical findings generated
10	User asks question in AI Chat	Response generated from analytical context
11	All agents complete	Final report generated

C. Results and Observations

The prototype successfully completed the intended end-to-end workflow from raw dataset upload to final report generation.

The Profiling Agent correctly identified dataset dimensions, column types, missing values and duplicate records.

The Cleaning Agent processed identified data-quality issues before downstream analysis.

The EDA Agent generated statistical summaries and exploratory findings.

When a suitable prediction problem was identified, the Modeling Agent trained and evaluated candidate models.

The Visualization Agent generated charts appropriate to the detected data characteristics.

The AI Insights Agent transformed analytical outputs into natural-language observations.

The final Report module consolidated the outputs into a structured report containing an executive summary, data-quality information, analysis and recommendations.

The modular architecture also allowed individual stages to be inspected independently, improving transparency compared with a single black-box analysis operation.

D. Comparison with Objectives

All primary system objectives were addressed:

- raw dataset upload is supported;
- automated profiling is performed;
- data cleaning is incorporated;
- exploratory analysis is automated;
- machine-learning modeling is supported when applicable;
- visualizations are generated automatically;
- AI-based insights are produced;
- conversational analysis is available; and
- the complete workflow is consolidated into a final report.

VII. DISCUSSION, LIMITATIONS AND FUTURE WORK

The main strength of Autonomous Data Scientist is its **end-to-end orchestration**. Instead of providing isolated data-analysis utilities, the system connects multiple analytical stages into a single autonomous workflow.

The agent architecture also provides modularity. Individual agents can be improved independently while maintaining the overall workflow.

Another advantage is the separation between computational analysis and AI interpretation. Numerical statistics, model metrics and visualizations are generated by analytical components and subsequently provided as context to the AI layer.

The main limitation is that complete automation of data science is difficult for arbitrary datasets. Correct analytical decisions frequently require domain knowledge that cannot be inferred from the dataset alone.

A second limitation is automated target identification. Not every dataset contains an obvious prediction target, and selecting an inappropriate target can produce misleading modeling results.

A third limitation is AI-generated interpretation. Although grounding insights in computed results reduces unsupported statements, generated explanations still require validation for high-stakes applications.

Further limitations include computational cost for large datasets and limited support for unstructured or multimodal data.

Future work includes:

Multi-agent collaboration: Multiple analytical agents can independently investigate a dataset and a reviewer agent can compare their findings.

Autonomous feature engineering: The system can generate and evaluate candidate features before modeling.

Model explainability: SHAP, feature importance and partial-dependence analysis can be incorporated.

Statistical validation: Statistical-test agents can validate automatically detected relationships.

Human-in-the-loop control: Users can approve or modify important analytical decisions.

Experiment tracking: Model configurations, datasets, metrics and results can be automatically recorded.

Multimodal analysis: Future versions can support text, images, audio and other data types.

Self-evaluation: A reviewer agent can verify whether generated insights are supported by the underlying statistical evidence.

VIII. CONCLUSION

Autonomous Data Scientist automates the end-to-end data-science workflow by coordinating specialised agents for profiling, cleaning, exploratory analysis, machine learning, visualization, insight generation and report creation. The

platform transforms a raw dataset into a structured analytical output while reducing repetitive manual operations.

The agent-based architecture provides a modular foundation for incorporating additional analytical capabilities without redesigning the complete system. The integration of AI-generated insights and conversational interaction further allows users to explore analytical results using natural language.

The current implementation demonstrates the feasibility of an autonomous workflow for structured tabular data. Future development will focus on multi-agent collaboration, autonomous feature engineering, model explainability, statistical validation, self-evaluation and human-in-the-loop decision mechanisms.

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